

Extending a homeomorphism

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We want to prove the following theorem on metric spaces $E, F, \bar{F}, \bar{E}$:

Theorem I. *If F is closed in E , then a homeomorphism between F and $\bar{F}$ can be extended to a homeomorphism between E and a suitable space $\bar{E}$.*

In other words, it asserts the possibility of “Re-metrization”. Instead of the old metric uv defined on E , we can introduce a new metric $\bar{u}\bar{v}$ defined on E which defines a homeomorphism E to $\bar{E}$ and for which the distance of the values of a pair of points in F of the homeomorphism is the same as the prescribed metric of $\bar{F}$ (homeomorphism means that the formula $uv \rightarrow 0$ with fixed u implies $\bar{u}\bar{v} \rightarrow 0$ and vice versa).

One can also say that a metrizable topological space E always can have a metric whose submetric is the prescribed one on a closed subset $F \subset E$.

The proof of Theorem I is similar to my previous proof (Math. Z. 5 (1919), p. 296) of the H. Tietze theorem, which states that a continuous function on a closed subset can be extended to a continuous one on the whole space, but it requires considerably more artifice.

We denote by u, v, w points in the space E , by a, b, c, p, q, r points in the closed space F , by x, y points in the open complement $G = E \setminus F$. Let

$$\delta(u) = \inf_p pu$$

be the infimum of the (old) distance of the point u and the point $p \in F$ (the minimum distance between u and F). Then $\delta(a) = 0$, $\delta(x) > 0$. From $pu \leq pv + uv$, by taking the infimum on p of both sides, we have

$$|\delta(u) - \delta(v)| \leq uv.$$

Since we will have to use them later, we now consider the function

$$(0) \quad \delta(u, v) = \min[uv, \delta(u) + \delta(v)],$$

which is symmetric and vanishes if and only if $u = v$ or two points u, v are in F . It satisfies the “triangle inequality”

$$\delta(u, v) + \delta(v, w) \geq \delta(u, w).$$

In fact, the four inequality hold

$$\begin{aligned}
uv + uw &\geq uw \\
uv + [\delta(v) + \delta(w)] &\geq \delta(u) + \delta(w) \\
[\delta(u) + \delta(v)] + vw &\geq \delta(u) + \delta(w) \\
[\delta(u) + \delta(v)] + [\delta(v) + \delta(w)] &\geq \delta(u) + \delta(w)
\end{aligned}$$

and if one takes the minimum of the right and left hand sides of the four formulas, then the triangle inequality holds.

Theorem I is equivalent to the following:

Theorem II. *If F is a closed set in E with a homeomorphism into $\overline{F}$, then there is a function $\varphi(u, a) \geq 0$ with the following properties:*

- (α) $\varphi(b, a) = \overline{ab}$
- (β) if $\delta(x) \rightarrow 0$ and $\varphi(x, a) \rightarrow 0$, then $ax \rightarrow 0$ (with fixed a)
- (γ) For each pair of points u, v , the supremum

$$(1) \quad \psi(u, v) = \sup_a |\varphi(u, a) - \varphi(v, a)|$$

is finite and converges to 0 when $uv \rightarrow 0$ (with fixed u).

In fact, if Theorem I is valid, then the new distance $\varphi(u, a) = \overline{au}$ satisfies the condition of Theorem II. Note that $\psi(u, v) \leq \overline{uv}$. The condition (β) holds since $\overline{ax} \rightarrow 0$ and $ax \rightarrow 0$ occur simultaneously, and the condition (γ) hold since $uv \rightarrow 0$ and $\overline{uv} \rightarrow 0$ occur simultaneously.

On the other hand, to transfer Theorem II to Theorem I, we first notice that $\psi(u, v)$ is symmetric and satisfies the triangle inequality

$$\psi(u, v) + \psi(v, w) \geq \psi(u, w).$$

This function is not yet usable as a new distance since it possibly vanishes for $u \neq v$.

Therefore, by using a function (0), we define

$$(2) \quad \overline{uv} = \max[\psi(u, v), \delta(u, v)],$$

and we show that this is a suitable distance. Anyway, $\overline{uv}$ is symmetric and satisfies the triangle inequality $\overline{uv} + \overline{vw} \geq \overline{uw}$.

Let us express the property (γ) separately depending on the three cases that u, v are points of F and F , or F and G , or G and G :

- (γ_1) $\psi(b, c) = \sup_a |\overline{ab} - \overline{ac}| = \overline{bc}$ holds
- (γ_2) $\psi(x, b) = \psi(b, x) = \sup_a |\varphi(x, a) - \overline{ab}| \geq \varphi(x, b)$ is finite and converges to 0 when $xb \rightarrow 0$ (with fixed b)¹
- (γ_3) $\psi(x, y) = \sup_a |\varphi(x, a) - \varphi(y, a)|$ is finite and converges to 0 when $xy \rightarrow 0$ (with fixed x).

¹ $xb \rightarrow 0$ with fixed x is out of question.

Now we see that $\overline{uv}$ vanishes if and only if $u = v$. Since $\overline{uv} = 0$ means that $\delta(u, v) = \psi(u, v) = 0$, either $u = v$, or $u = b$ and $v = c$ (where $b, c \in F$) holds, but in the second case, $\psi(b, c) = \overline{bc} = 0$, then we have $b = c$. According to this, $\overline{uv}$ provides a distance on a space E , and it remains to prove that the space with this new distance is homeomorphic to the original space E .

If $uv \rightarrow 0$ (with fixed u), then $\delta(u, v) \rightarrow 0$. By (γ) , we have $\psi(u, v) \rightarrow 0$. Thus $\overline{uv} \rightarrow 0$.

If $\overline{uv} \rightarrow 0$ (with fixed u), then we have $\delta(u, v) \rightarrow 0$ and $\psi(u, v) \rightarrow 0$. If $u = x$ (where $x \in G$), then from $\delta(x, v) = \min[xv, \delta(x) + \delta(v)]$ and $\delta(x) > 0$ and $\delta(x, v) \rightarrow 0$ it follows that $xv \rightarrow 0$. If $u = b$ ($b \in F$), then we have $\delta(b, v) = \min[bv, \delta(v)] = \delta(v) \rightarrow 0$ and $\psi(b, v) \rightarrow 0$. Here the cases $v = c \in F$ and $v = y \in G$ is distinguished (at least one of which infinitely often must occur when a variable v moves). In the first case, $\psi(b, c) = \overline{bc} \rightarrow 0$ and $bc \rightarrow 0$. In the second case, $\delta(y) \rightarrow 0$ and $\psi(b, y) \rightarrow 0$ and even more we have $\varphi(y, b) \rightarrow 0$. Due to (β) , we have $by \rightarrow 0$. Therefore $\overline{uv} \rightarrow 0$ always implies $uv \rightarrow 0$.

This shows that Theorem I follows from Theorem II. It is now a matter of construction of a function $\varphi(u, a)$, which satisfies the properties (α) (β) (γ) , or a function $\varphi(x, a)$, which satisfies the properties (β) (γ_2) (γ_3) .

We define (taking the infimum and supremum on $p \in F$) the infimum

$$(3) \quad \omega(x, c) = \inf_p \left[\overline{cp} + 2 \frac{px}{\delta(x)} - 2 \right]$$

and the supremum

$$(4) \quad \varphi(x, a, c) = \sup_p \left[\overline{ap} - \overline{cp} - \frac{px}{\delta(x)} + 1 \right],$$

which (even if F is unbounded) is certainly finite since bracketed formula is $\leq \overline{ac}$. Then we have

$$-\varphi(x, a, c) = \inf_p \left[\overline{cp} - \overline{ap} + \frac{px}{\delta(x)} - 1 \right] \leq \omega(x, c)$$

and thus,

$$(5) \quad \varphi(x, a) = \varphi(x, a, c) + \omega(x, c) \geq 0,$$

where c is any fixed point in F . The proof that $\varphi(x, a)$ has the required properties follows from the following consideration.

(A) We choose a point p with $px < \delta(x) + \delta(x)^2$, then $(px/\delta(x)) - 1 < \delta(x)$. Thus, we have

$$\begin{aligned} \omega(x, c) &< \overline{cp} + 2\delta(x) \\ \varphi(x, a, c) &> \overline{ap} - \overline{cp} - \delta(x). \end{aligned}$$

When $bx \rightarrow 0$ with fixed b , we have $\delta(x) \rightarrow 0$, $px \rightarrow 0$, and $bp \rightarrow 0$. Thus we have $\overline{bp} \rightarrow 0$, $\overline{ap} \rightarrow \overline{ab}$, and $\overline{cp} \rightarrow \overline{cb}$, and we also conclude

that when $bx \rightarrow 0$, we have

$$\overline{\lim} \omega(x, c) \leq \overline{cb}, \quad \underline{\lim} \varphi(x, a, c) \geq \overline{ab} - \overline{cb}.$$

(B) We choose a point q dependent to x (and c) with

$$(6) \quad \omega(x, c) + \delta(x) > \overline{cq} + 2 \frac{qx}{\delta(x)} - 2.$$

On the other hand,

$$\omega(x, c) \leq \overline{cb} + 2 \frac{bx}{\delta(x)} - 2.$$

By combining these, we obtain

$$2 \frac{qx}{\delta(x)} < \overline{cb} + 2 \frac{bx}{\delta(x)} + \delta(x)$$

$$2qx < 2bx + \overline{cb} \cdot \delta(x) + \delta(x)^2$$

From these, it follows again that if $bx \rightarrow 0$ (with fixed b), then we have $qx \rightarrow 0$, $bq \rightarrow 0$, $\overline{bq} \rightarrow 0$ and $\overline{cq} \rightarrow \overline{cb}$. From (6) or $\omega(x, c) > \overline{cq} - \delta(x)$ it follows that

$$\underline{\lim} \omega(x, c) \geq \overline{cb}.$$

By combining this inequality with (A), we conclude that when $bx \rightarrow 0$, we have $\omega(x, c) \rightarrow \overline{cb}$.

Next we have

$$\varphi(x, a, c) \geq aq - \overline{cq} - \frac{qx}{\delta(x)} + 1,$$

By adding this to (6), we have

$$\varphi(x, a) + \delta(x) > \overline{aq} + \frac{qx}{\delta(x)} - 1.$$

This contains the proof of (β) since $\varphi(x, a) \rightarrow 0$ with fixed a and $\delta(x) \rightarrow 0$ imply $\overline{aq} \rightarrow 0$ and $(qx/\delta(x)) - 1 \rightarrow 0$. Hence we have $aq \rightarrow 0$, $qx \rightarrow 0$, and $ax \rightarrow 0$.

Finally from $\varphi(x, a) > \overline{aq} - \delta(x)$ we conclude that

$$\overline{ab} - \varphi(x, a) < \overline{ab} - \overline{aq} + \delta(x) \leq \overline{bq} + \delta(x).$$

Hence one half of (γ_2) is proven since the positive bounded $\overline{bq} + \delta(x)$ of $\overline{ab} - \varphi(x, a)$ is dependent a (q depended only on x and any fixed c). Thus the function $\overline{bq} + \delta(x)$ (dependent only x and b) converges to 0 when $bx \rightarrow 0$, since we have already found $\overline{bq} \rightarrow 0$.

(C) We choose a point r dependent on x , a (and c) with

$$(7) \quad \begin{aligned} \varphi(x, a, c) - \delta(x) &< \overline{ar} - \overline{cr} - \frac{rx}{\delta(x)} + 1 \\ &\leq \overline{ac} - \frac{rx}{\delta(x)} + 1. \end{aligned}$$

On the other hand, we have

$$\begin{aligned}\varphi(x, a, c) &\geq \overline{ab} - \overline{cb} - \frac{bx}{\delta(x)} + 1 \\ &\geq -\overline{ac} - \frac{bx}{\delta(x)} + 1.\end{aligned}$$

By combining these inequalities, we obtain

$$\begin{aligned}\frac{rx}{\delta(x)} &< 2\overline{ac} + \frac{bx}{\delta(x)} + \delta(x), \\ rx &< bx + 2\overline{ac} \cdot \delta(x) + \delta(x)^2.\end{aligned}$$

From these, it follows again that if $bx \rightarrow 0$ with fixed b , then we have $rx \rightarrow 0$, $br \rightarrow 0$, $\overline{br} \rightarrow 0$, $\overline{ar} \rightarrow \overline{ab}$, and $\overline{cr} \rightarrow \overline{cb}$. From (7) or $\varphi(x, a, c) < \overline{ar} - \overline{cr} + \delta(x)$, we conclude that $\lim \varphi(x, a, c) \leq \overline{ab} - \overline{cb}$. By combining this inequality with (A), we conclude that when $bx \rightarrow 0$, we have $\varphi(x, a, c) \rightarrow \overline{ab} - \overline{cb}$.

When $bx \rightarrow 0$, the valid limit equations

$$\omega(x, c) \rightarrow \overline{cb}, \quad \varphi(x, a, c) \rightarrow \overline{ab} - \overline{cb}$$

follows. Then, by adding these, we have $\varphi(x, a) \rightarrow \overline{ab}$, in particular, $\varphi(x, b) \rightarrow 0$.

(D) By adding

$$-\overline{cp} - \frac{px}{\delta(x)} + 1$$

to both sides of $\overline{ap} \leq \overline{bp} + \overline{ab}$, and by taking the supremum on p , we have

$$\varphi(x, a, c) \leq \varphi(x, b, c) + \overline{ab}.$$

By adding $\omega(x, c)$ to this inequality, we have

$$\varphi(x, a) \leq \varphi(x, b) + \overline{ab}.$$

This is the second half of the assertion (γ_2) since, similarly to $\overline{ab} - \varphi(x, a)$, now also $\varphi(x, a) - \overline{ab}$ has the upper bound $\varphi(x, b)$ independent of a , and depending only on x and b , and which is ≥ 0 , namely $\varphi(x, b)$ converges to 0 when $bx \rightarrow 0$.

(E) In order to prove (γ_3), we note that

$$px \cdot \delta(y) - py \cdot \delta(x) = [px - py]\delta(y) + py[\delta(y) - \delta(x)],$$

and its absolute value is $\leq xy \cdot [\delta(y) + py] \leq 2xy \cdot py$. Now, without loss of generality, we can assume that E is bounded, for example, which diameter is $\leq 1/2$, since each space is homeomorphic to a bounded one (one replace the distance δ with $\delta/1 + \delta$, preserving the triangle inequality). On the other hand, it would not be permissible to assume the set $\overline{F}$ to be bounded without reducing the scope of Theorem I). Then we have

$$\left| \frac{px}{\delta(x)} - \frac{py}{\delta(y)} \right| \leq \frac{xy}{\delta(x)\delta(y)},$$

and according to the same method in (D) we have

$$\begin{aligned} |\omega(x, c) - \omega(y, c)| &\leq \frac{2xy}{\delta(x)\delta(y)}, \\ |\varphi(x, a, c) - \varphi(y, a, c)| &\leq \frac{xy}{\delta(x)\delta(y)}, \\ |\varphi(x, a) - \varphi(y, a)| &\leq \frac{3xy}{\delta(x)\delta(y)}, \end{aligned}$$

where the upper bounds on the right hand side independent of a converge to 0 when $xy \rightarrow 0$ with fixed x (note that $\delta(x) \rightarrow \delta(y) > 0$).

This completes the proof of Theorem I.

Simple consequences of Theorem I are the the following two statements, of which the second one was proven by V. Niemytzki und A. Tychonoff in a different way (Fund. Math. 12 (1928), pp. 118–120):

- (1) *A metrizable space is compact if and only if it is bounded for every metric generating the same topology.*
- (2) *A metrizable space is compact if and only if it is complete for every metric generating the same topology.*

The necessity of the conditions is clear. On the other hand, if E is not compact, then it contains a closed countable set $F = \{a_1, a_2, \dots, a_n, \dots\}$ without an accumulation point. Since two discrete sets of equal cardinality are always homeomorphic to each other, F is homeomorphic to both of

$$F_1 = \{1, 2, \dots, n, \dots\}$$

and

$$F_2 = \left\{1, \frac{1}{2}, \dots, \frac{1}{n}, \dots\right\}.$$

According to Theorem I, E is homeomorphic to an unbounded space E_1 and to a non-complete space E_2 (in which F_2 is closed, i.e., a fundamental sequence without limit exists).

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